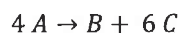
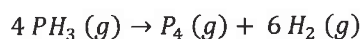


## CHEMICAL REACTION ENGINEERING (CHE 321)

## MIDTERM EXAM 1

CLOSED BOOK, TIME 70 MIN

The homogeneous gas-phase decomposition of phosphine



proceeds at 922.15 K with the first-order rate:

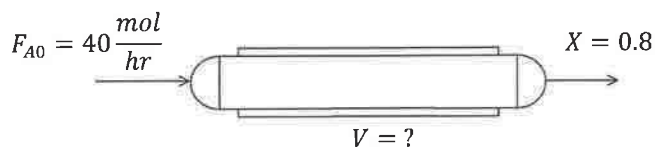
$$r_A = -k C_A$$

$$k = 10 \text{ hr}^{-1}$$

- 45 1. What size of plug flow reactor (PFR), operating at constant temperature and pressure of 922.15 K and 460 kPa, can achieve 80% conversion of a feed of 40 moles of pure phosphine per hour?

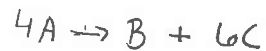
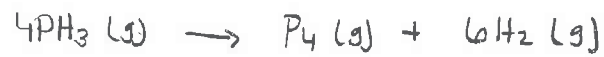
Conversion  $X \equiv \frac{F_{A0} - F_A}{F_{A0}}$ . Gas constant  $R = 8.314 \frac{\text{Pa.m}^3}{\text{mol.K}}$ .  $\int dx \frac{1+\epsilon x}{1-x} = -(1+\epsilon) \ln(1-x) - \epsilon x$ .

- 5 1.1. Based on a shell mole balance, derive a differential equation for conversion as a function of PFR volume.  
 1.2. Express the rate of reaction in terms of conversion  $X$ .  
 1.3. Integrate the differential mole balance and determine the volume.



Name: Bushra Alkarmi

#1



$$-r_A = -kC_A$$

$$k = 16 \text{ hr}^{-1}, \quad F_{A0} = 40 \text{ mol/hr}, \quad T = 922.15 \text{ K}, \quad P = 460 \text{ kPa}, \quad x = 0.8$$

$$1.1: \quad \frac{dF_A}{dV} = r_A, \quad F_A = F_{A0}(1-x)$$

$$\frac{dF_A}{dV} = -F_{A0} \frac{dx}{dV}$$

$$-F_{A0} \frac{dx}{dV} = r_A \quad \rightarrow \quad \frac{dx}{dV} = \frac{-r_A}{F_{A0}}$$

$$V = F_{A0} \int_0^x \frac{dx}{-r_A}$$

$$1.2 \quad -r_A = kC_A$$

$$C_A = C_{A0} \frac{1-x}{1+\epsilon x}$$



$$\Delta V = (1+6) - 4 = 3$$

$$\epsilon = \frac{\Delta V}{V_A}$$

$$= \frac{3}{4}$$

$$\bar{\epsilon} = 0.75$$

Name:

#2  $k = 10 \text{ hr}^{-1}$ ,  $F_{A0} = 40 \text{ mol/hr}$ ,  $X = 0.8$ ,  $V = 250 \text{ dm}^3 = 250 \text{ L}$   
 $T = 649^\circ\text{C} = 922.15 \text{ K}$ ,  $P = 230 \text{ kPa}$ ,  $R = 8.314$ ,  $\epsilon = 0.75$

2.1

$$V_2 = \frac{F_{A0} (X_2 - X_1)}{-r_A}$$

2.2

$$C_A = C_{A0} \frac{1-X}{1+\epsilon X}$$

$$C_{A0} = \frac{P}{RT} = \frac{230}{(8.314)(922.15)} = 0.030 \text{ mol/L}$$

$$\rightarrow -r_A = k C_{A0} \frac{1-X_2}{1+\epsilon X_2}$$

$$-r_A = (10)(0.030) \frac{1-X_2}{1+0.75X_2}$$

$$-r_A = 0.30 \frac{1-X_2}{1+\epsilon X_2}$$